

Student Number:



2008

Mid Year Examination

HSC CHEMISTRY

Thursday 3 April, 9 – 11am

General Instructions

- Reading time - 5 minutes
- Working time - 2 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board approved calculators may be used
- A Data Sheet and Periodic Table are provided at the back of this paper
- Write your student number at the top of every page.

Mr Hunter
Mr Lee
Mr Weeding
40 students

Total marks - 65

Section I – 45 marks

Part A (15 marks)

- Attempt Questions 1-15

Part B (30 marks)

- Attempt Questions 16- 20

Section II - 20 marks

- Attempt Question 21

Student Number:

Question 15

Source: Smith, R. (2005). Conquering Chemistry. McGraw-Hill Australia.

Section 1

45 marks

Part A**Total marks 15****Attempt questions 1-15**

Use the Multiple-choice Answer Sheet provided.

Answer the questions by selecting the alternative that best answers the question. Indicate your choice by filling in the appropriate place on the Answer sheet, as shown below, where **A** has been selected as the best alternative,

A ☒ **B** ☐ **C** ☐ **D** ☐

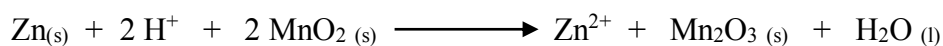
If you make a mistake, indicate your choice by labelling the correct alternative, as shown below where, the original choice **A** was a mistake, and **C** is now selected as being the correct answer.

A ☒ ~~X~~ **B** ☐ **C** ☒ ^{Correct} **D** ☐

1 Which of the following statements correctly describes a redox reaction?

- (A) The oxidation half-reaction and the reduction half-reaction occur simultaneously.
- (B) The oxidation half-reaction occurs before the reduction half-reaction.
- (C) The oxidation half-reaction occurs after the reduction half-reaction.
- (D) The oxidation half-reaction occurs spontaneously but the reduction half-reaction does not.

2. Consider the following reaction



What is the oxidant in this reaction?

- (A) $\text{MnO}_{2(s)}$
- (B) $\text{Mn}_2\text{O}_{3(s)}$
- (C) $\text{Zn}_{(s)}$
- (D) H^+

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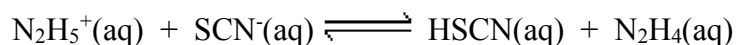
3. The hydronium ion concentration (in mol L^{-1}) of some common substances is given in the Table below

$[\text{H}_3\text{O}^+]$	Substance
10^{-9}	baking soda
10^{-5}	black coffee
10^{-8}	sea water
10^{-11}	laundry detergent
10^{-6}	milk
10^{-13}	chlorine bleach
10^{-4}	soda water

Of the substances listed which of the following are acidic?

- (A) Soda water and chlorine bleach
- (B) Milk and laundry detergent
- (C) Sea water and baking soda
- (D) Black coffee and milk

4. In the equilibrium

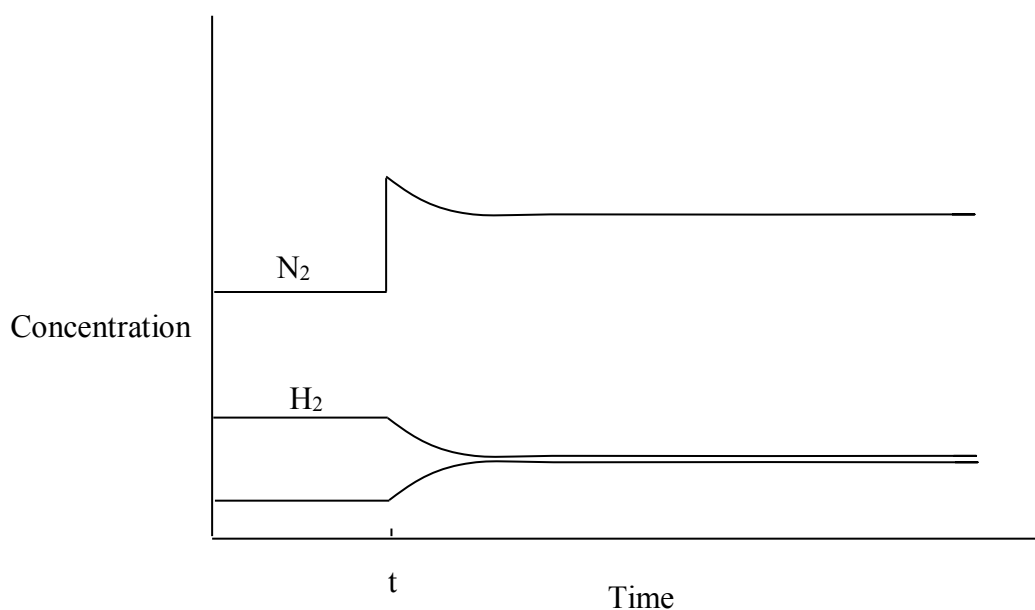


- (A) N_2H_5^+ acts as a acid
- (B) SCN^- acts as a acid
- (C) HSCN acts as a base
- (D) N_2H_4 acts as a acid

5. Which statement best describes a weak acid solution?

- (A) There are no neutral acid molecules present.
- (B) Only a fraction of the acid molecules is ionised.
- (C) All acid present is ionised to hydrogen ions.
- (D) The total concentration of acid molecules present is high.

6. In the process of esterification the reactant alcohol and acid mixture is refluxed. What is the purpose of refluxing the mixture?
- (A) Speed up the reaction
 - (B) Prevent the loss of alcohol as the reactant mixture is heated
 - (C) Remove the water produced as a product of the reaction
 - (D) Force the reaction to come to equilibrium
7. An equilibrium mixture between nitrogen, hydrogen and ammonia was subjected to a change at time t . The result of this change is shown in the diagram below.



What was the change made to the equilibrium mixture at time t ?

- (A) The pressure in the equilibrium mixture of nitrogen and hydrogen was decreased.
- (B) The temperature of the reaction mixture was raised.
- (C) The concentration of nitrogen gas in the equilibrium mixture was increased.
- (D) The volume of the reaction vessel was increased.

8. Which substance can act both as an acid and as a base, in dilute solutions?

- (A) Calcium carbonate
- (B) Ammonium nitrate
- (C) Ethanol
- (D) Water

9. The pH values of four acids and their concentrations are shown in the table below.

Acid	Conc. (mol L ⁻¹)	pH
P	0.01	2.0
Q	0.05	1.0
R	0.1	1.0
S	0.1	2.0

Which acid can donate more than one proton?

- (A) P
- (B) Q
- (C) R
- (D) S

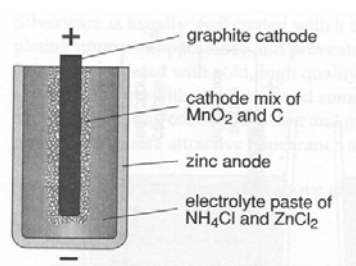
10. Which group of substances below result in a lower pH when dissolved in water?

- (A) Ammonia, sodium hydroxide, potassium carbonate
- (B) Hydrogen chloride, ethanol, carbon monoxide
- (C) Sodium oxide, magnesium oxide, calcium hydroxide
- (D) Carbon dioxide, sulfur dioxide, hydrogen bromide

11. When making the ester, ethyl propanoate, concentrated sulfuric acid is added to a mixture of ethanol and propanoic acid. One effect of the sulfuric acid is to increase the yield of the ester. Which of the following is the correct explanation for this increased yield?

- (A) Sulfuric acid is a dehydrating agent and removes water as a reaction product.
- (B) Sulfuric acid provides hydrogen ions which catalyse the reaction.
- (C) The mixture becomes hot, which accelerates the reaction.
- (D) The boiling point of the mixture increases, allowing a higher reaction temperature.

12. Which of the following 4 carbon atom molecules has the highest boiling point?
- (A) Butane
 - (B) Methyl propanoate
 - (C) Butanoic acid
 - (D) 2-butanol
13. A 285.3 g bottle of soda awater was deccarboanted by adding 14.7 g of salt into it. The final mass was 296 g. The volume of gas formed at 25°C and 100kPa was -
- (A) 2.22 L
 - (B) 2.06 L
 - (C) 2.25 L
 - (D) 2.04 L
14. When an indicator is used in acid base titration, its colour change -
- (A) always identifies the equivalence point.
 - (B) only occurs at a pH of 7.
 - (C) needs to correspond to that point when the moles of acid and base present in the mixture are identical.
 - (D) needs to correspond to that point when the moles of acid and base present in the mixture correspond to the molar ratio of the balanced equation.
15. The diagram below shows a dry cell battery.



Which of the following statements is correct for this dry cell battery?

- (A) The manganese(IV) oxide is the electrolyte.
- (B) The graphite rod is the anode
- (C) Graphite is reduced at the cathode
- (D) Zinc is oxidised to zinc(II) at the anode.

Section 1 (continued)

Part B – 30 marks

Attempt Questions 16-20

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 16 (10 marks)

Marks

- (a) (i) From the Table of Standard Potentials, select TWO metals that will reduce hydrogen ions to hydrogen gas.

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- (ii) Using one of the metals in part (i) write oxidation and reduction half reactions for the reaction which occurs.

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- (iii) Write a balanced overall cell equation for the redox reaction.

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Question 16 continues on page 9.

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|-----|--|---|--------------|
| (b) | | (i) Sketch and label a diagram showing the structure of ONE of the following cells: | Marks |
| | | <ul style="list-style-type: none">• button cell• fuel cell• lithium cell• liquid junction photovoltaic device (eg the Gratzel cell),• vanadium redox cell | 3 |

- | | | | |
|--|--|--|----------|
| | | (ii) Explain the chemistry of the cell drawn in b(i) | 3 |
|--|--|--|----------|

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Question 17 (9 marks)

- (a) (i) Define acids and bases according to the Bronsted-Lowry theory.

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- (ii) In the following two reactions, state whether HCO_3^- behaves as an acid or a base. Explain your answer in each case.

2

1. $\text{H}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$
2. $\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$

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- (b) The pH of a 0.001 mol L^{-1} solution of hydrochloric acid and the pH of a 0.056 mol L^{-1} of acetic (ethanoic, CH_3COOH) acid is 3.

- (i) Compare the concentration of each acid. Explain your answer.

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- (ii) Compare the strength of each acid. Explain your answer.

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Question 17 continues on page 11.

Student Number:

Marks

(iii) Compare the hydrogen ion concentration in the solutions of each acid. Explain your answer.

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Question 18 (6 marks)

During your practical course you have performed a first hand investigation whereby you made an ester.

- (a) Identify the ester produced. **1**

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- (b) Write a balanced chemical equation for the production of this ester. **1**

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- (c) Justify the procedure used to produce this ester. **4**

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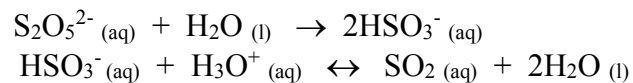
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Question 19 (2 marks)

Marks

Potassium metabisulphite, $\text{K}_2\text{S}_2\text{O}_5$, is added to wine to prevent oxidation. The reaction mechanism is as follows:



Use the equations to explain how adding potassium metabisulphite will affect the pH of wine.

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Student Number:

Marks

Question 20 (3 marks)

- (a) Use an equation to show the ionisation of acetic (ethanoic) acid in water.

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- (b) Use a diagram to model the ionisation of acetic (ethanoic) acid in water.

2

Section II**20 marks****Answer Question 21 below.**

Answer the question in a writing booklet. Extra writing booklets are available. Show all relevant working in questions involving calculations.

Question 21 – Shipwrecks, Corrosion and Conservation (20 marks)		Marks
(a)	A solution of copper(II) chloride is electrolysed using inert graphite electrodes. At one of the electrodes, bubbles form. Analysis of these bubbles shows that two different gases are being formed, one of which is confirmed to be chlorine.	
(i)	In your answer booklet, draw a fully labelled diagram of the equipment used for this electrolysis.	2
(ii)	With the use of a half equation, identify the other gas produced.	1
(iii)	Calculate the applied voltage required for this electrolysis reaction to proceed.	2
(iv)	As the reaction proceeds, other observable changes occur. Describe two of these changes and explain why they occur.	2
(b)	(i) Identify the trend in oxygen concentration with increasing depth in the ocean.	1
	(ii) Explain why the oxygen concentration changes with increasing depth.	2
(c)	The work of Galvani, Volta, Davy and Faraday has helped in increasing an understanding of electron transfer reactions. Describe the impact of this work on society.	4
(d)	You performed a first-hand investigation to compare and describe the rate of corrosion of metals in different acidic and neutral solutions.	
(i)	Outline the procedure used and the results you obtained.	2
(ii)	Identify a risk associated with this procedure.	1
(iii)	Use your results to explain why metal corrosion is accelerated in an acidic environment.	3

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DATA SHEET

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and

at 0°C (273.15 K) 22.71 L

at 25°C (298.15 K) 24.79 L

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

KEY		Atomic Number	Symbol of element	Name of element
79	Au	197.0	Gold	
5	B	10.81	Boron	
6	C	12.01	Carbon	
7	N	14.01	Nitrogen	
8	O	16.00	Oxygen	
9	F	19.00	Fluorine	
10	Ne	20.18	Neon	
11	Na	22.99	Sodium	
12	Mg	24.31	Magnesium	
13	Al	26.98	Aluminium	
14	Si	28.09	Silicon	
15	P	30.97	Phosphorus	
16	S	32.07	Sulfur	
17	Cl	35.45	Chlorine	
18	Ar	39.95	Argon	
19	K	39.10	Potassium	
20	Ca	40.08	Calcium	
21	Sc	44.96	Scandium	
22	Ti	47.87	Titanium	
23	V	50.94	Vanadium	
24	Cr	52.00	Chromium	
25	Mn	54.94	Manganese	
26	Fe	55.85	Iron	
27	Co	58.93	Cobalt	
28	Ni	58.69	Nickel	
29	Cu	63.55	Copper	
30	Zn	65.39	Zinc	
31	Ga	69.72	Gallium	
32	Ge	72.61	Germanium	
33	As	74.92	Arsenic	
34	Se	78.96	Selenium	
35	Br	79.90	Bromine	
36	Kr	83.80	Krypton	
37	Rb	85.47	Rubidium	
38	Sr	87.62	Strontium	
39	Y	88.91	Yttrium	
40	Zr	91.22	Zirconium	
41	Nb	92.91	Niobium	
42	Mo	95.94	Molybdenum	
43	Tc	[98.91]	Technetium	
44	Ru	101.1	Ruthenium	
45	Rh	102.9	Rhodium	
46	Pd	106.4	Palladium	
47	Ag	107.9	Silver	
48	Cd	112.4	Cadmium	
49	In	114.8	Indium	
50	Sn	118.7	Tin	
51	Sb	121.8	Antimony	
52	Te	127.6	Tellurium	
53	I	126.9	Iodine	
54	Xe	131.3	Xenon	
55	Cs	132.9	Cesium	
56	Ba	137.3	Barium	
57-71	Lanthanides			
72	Hf	178.5	Hafnium	
73	Ta	180.9	Tantalum	
74	W	183.8	Tungsten	
75	Re	186.2	Rhenium	
76	Os	190.2	Osmium	
77	Ir	192.2	Iridium	
78	Pt	195.1	Platinum	
79	Au	197.0	Gold	
80	Hg	200.6	Mercury	
81	Tl	204.4	Thallium	
82	Pb	207.2	Lead	
83	Bi	209.0	Bismuth	
84	Po	[210.0]	Polonium	
85	At	[210.0]	Astatine	
86	Rn	[222.0]	Radon	
87	Fr	[223.0]	Francium	
88	Ra	[226.0]	Radium	
89-103	Actinides			
104	Rf	[261.1]	Rutherfordium	
105	Db	[262.1]	Dubnium	
106	Sg	[263.1]	Seaborgium	
107	Bh	[264.1]	Bohrium	
108	Hs	[265.1]	Hassium	
109	Mt	[268]	Meitnerium	
110	Uun	—	Ununium	
111	Uuu	—	Ununium	
112	Uub	—	Unubium	
113	Uuh	—	Ununium	
114	Uuq	—	Ununquadium	
115	Uup	—	Ununpentium	
116	Uuh	—	Ununhexium	
117	Uus	—	Ununseptium	
118	Uuo	—	Ununoctium	

Lanthanides

57	La	138.9	Lanthanum
58	Ce	140.1	Cerium
59	Pr	140.9	Praseodymium
60	Nd	144.2	Neodymium
61	Pm	[146.9]	Promethium
62	Sm	150.4	Samarium
63	Eu	152.0	Europium
64	Gd	157.3	Gadolinium
65	Tb	158.9	Terbium
66	Dy	162.5	Dysprosium
67	Ho	164.9	Holmium
68	Er	167.3	Erbium
69	Tm	168.9	Thulium
70	Yb	173.0	Ytterbium
71	Lu	175.0	Lutetium

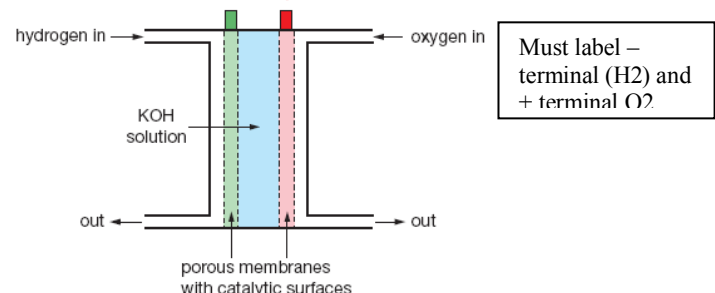
Actinides

89	Ac	[227.0]	Actinium
90	Th	232.0	Thorium
91	Pa	231.0	Protactinium
92	U	238.0	Uranium
93	Np	[237.0]	Neptunium
94	Pu	[239.1]	Plutonium
95	Am	[241.1]	Americium
96	Cm	[244.1]	Curium
97	Bk	[249.1]	Berkelium
98	Cf	[252.1]	Californium
99	Es	[252.1]	Einsteinium
100	Fm	[257.1]	Fermium
101	Md	[258.1]	Mendelevium
102	No	[259.1]	Nobelium
103	Lr	[262.1]	Lawrencium

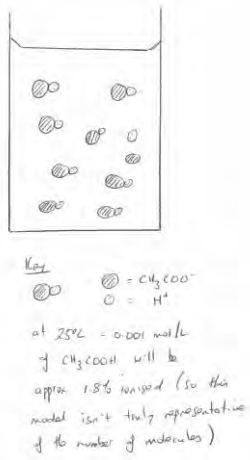
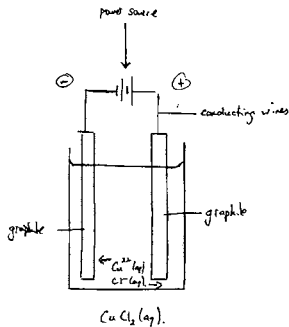
Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁸Tc.

**2008 HSC Half Year
Marking Guidelines**

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	A	D	A	B	B	C	D	B	D	A	C	C	D	D

Quest	Sample Answer	Marking Scheme
16 (a) (i)	any TWO of Mg, Al, Zn, Fe, Ni, Sn or Pb	Must have 2 for 1 mark
16 (a) (ii)	Oxidation half reaction $\text{Mg(s)} \leftrightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{e}^-$ Reduction half reaction $2\text{H}^{+}(\text{aq}) + 2\text{e}^- \leftrightarrow \text{H}_2(\text{g})$	1 for each correct oxidation and reduction equation
16 (a) (iii)	Redox cell reaction $\text{Mg} + 2\text{H}^{+} \leftrightarrow \text{Mg}^{2+} + \text{H}_2$	Correct combination of half cell reactions to overall redox cell reaction (depending on metal chosen) with states.
16(b) (i)		Correct sketch that represents accurate diagram of cell, correct labelling of electrodes, electrolytes, and terminals (-1 each error) NB Identify, by name of the cell chosen
16(b) (ii)	Cause: $\text{H}_2 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 4\text{e}^-$ Effect: The oxidation reaction involving hydrogen gas and the hydroxide ion provides the electrons for the reaction. Cause: $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$ Effect: The reduction reaction results due to the electrons provided at the anode to produce hydroxide ions. These reactions above produce water. $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ - Description to include at least an understanding of electrodes (porous carbon with a catalyst surface) and the fact they don't undergo change with electrode reactions - the cell can work indefinitely as long as fuel is supplied - electrolyte, gets diluted and must be replenished	<u>3 Marks</u> Correct half equations and correct net equation (1 mark) and description of the reactions (2 marks)
17 (a) (i)	An acid is a substance, that in solution, tends to <u>donate protons</u> , and a base is a substance that tends to <u>accept protons</u> .	1 all correct
17 (a) (ii)	<u>Cause:</u> the hydrogen carbonate ion is amphiprotic. <u>Effect:</u> In the first reaction, HCO_3^- is acting as a base as it accepts H^+ to form H_2CO_3; in the second reaction, HCO_3^- is acting as an acid as it donated a proton to form CO_3^{2-}.	1 mark each reaction

17 (b) (i)	Cause: The concentration of an acid refers to the amount of solute in a volume of solution Effect: Thus hydrochloric acid is less concentrated as 0.001 is lower in conc than 0.56	1 mark qualitative 2 marks quantitative
17 (b) (ii)	Cause: The strength of an acid refers to the degree of ionisation or dissociation in solution. Effect: Ethanoic is less ionised in solution and thus is the weaker acid. Using this information, it is only 1.8% ie $(0.001 \div 0.056) \times 100$ ionised while HCl is 100% ionised.	2 marks must be quantitative with working
17 (b) (iii)	Cause: Since the pH, which is a measure of hydrogen ion concentration, Effect: Each acid is the same, the hydrogen ion concentration in solution must be the same. $[H^+] = 10^{-pH}$ $= 10^{-3}$ $= 0.001 \text{ mol/L}$	2 marks must be quantitative with working with the correct explanation
18 (a)	(a) Ethyl ethanoate	1 mark – identifies the ester made.
18 (b)	* $CH_3COOH + CH_3CH_2OH \leftrightarrow CH_3COOCH_2CH_3 + H_2O$ * concentrated sulphuric acid + heat under reflux.	1 marks – balanced equation
18 (c)	Add together 5mL of ethanol, 6mL of ethanoic acid and 8 drops of sulphuric acid in a quick fit reaction flask that can be connected to a water cooled condensor. Justification: The concentrated sulphuric acid acts as a catalyst and speeds up the rate of reaction. It also absorbs water therefore drives the above equilibrium to the right and increases the yield of ester. Add a couple of glass boiling chips. Justification: Dispersal of heat to ensure even heating and prevents bumping. Heat in a hot water bath under reflux for 20 minutes. Justification: The hot water bath provides a constant heating temperature of 100°C around the flask which also ensures even heating. This increases the rate of reaction and results in a higher yield.. A Bunsen flame used directly would be too hot and as the reactants are flammable and volatile they maybe ignited by the flame. Reflux ensures that as reactants evaporate, they then get condensed and return to the reaction flask. After refluxing for 20 minutes distill and collect the fraction that evaporates at 75-85°C. Justification: In the reaction flask is a mixture of ethyl ethanoate, ethanol, ethanoic acid and sulphuric acid. Distillation helps separate the ester as this is the temperature it evaporates. Pour distillate into a separating funnel and add sodium carbonate until gas stops evolving. Justification: The sodium carbonate reacts with left over ethanoic acid and sulphuric acid that escaped into the distillate. The sodium carbonate reacts to produce salt,	<u>3-4 marks</u> - detailed description of necessary equipment and the refluxing process or suitable diagram. - Justification given for all equipment, chemicals (including catalyst, must say conc sulphuric acid) and refluxing. <u>2 marks</u> - Describes equipment and refluxing process or draws a diagram. - Identifies some chemicals and gives at least one reason for a piece of equipment. <u>1 mark</u> Outlines some chemicals or equipment

	carbon dioxide and water. Example: $\text{CH}_3\text{COOH} + \text{Na}_2\text{CO}_3 \leftrightarrow \text{CH}_3\text{COO}^- \text{Na}^+ + \text{CO}_2 + \text{H}_2\text{O}$ Separate by opening the tap of the funnel and release the aqueous layer. Justification: The oily layer is on top and is the ester. After releasing the aqueous layer, all that is left in the separating funnel is the ester.	
19	The addition of metabisulphite ($\text{S}_2\text{O}_5^{2-}$) increases the concentration of HSO_3^- as shown in the first equation. The second equation shows an equilibrium reaction. The system adjusts to overcome this increase in the concentration of HSO_3^- by favouring the forward reaction. This causes the concentration of H_3O^+ to decrease and the pH to increase.	3-4 marks Clearly demonstrates a quantitative understanding of pH as it relates to $[\text{H}_3\text{O}^+]$ AND explains the effect of a decrease in $[\text{H}_3\text{O}^+]$ concentration with reference to the second equation. 2 marks Demonstrates an understanding of pH as it relates to $[\text{H}_3\text{O}^+]$ AND explains the effect of a decrease in $[\text{H}_3\text{O}^+]$ concentration with reference to the second equation. 1 mark Identifies the pH change as becoming more basic or less acidic
20 (a)	$\text{CH}_3\text{COOH} (\text{aq}) \leftrightarrow \text{CH}_3\text{COO}^- (\text{aq}) + \text{H}^+ (\text{aq})$	Must have states and equil arrow.
20 (b)		Must be quantitative
21	(i) 	2 Marks - Must have electrodes labelled, solutions labelled, electron flow, polarity of electrodes. Subtract 1 for each omission. 1 Mark - Correct identification of gas.

	(ii) $2\text{H}_2\text{O(l)} \rightarrow \text{O}_2\text{(g)} + 4\text{H}^+\text{(aq)} + 4\text{e}^-$ Gas produced is oxygen gas (iii) $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu} \quad +0.34\text{V}$ $\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^- \quad -1.36\text{V}$ $\text{Cu}^{2+} + 2\text{Cl}^- \rightarrow \text{Cu} + \text{Cl}_2 \quad -1.02\text{V}$ So a voltage <u>greater than 1.02V</u> must be applied. (iv) The negative electrode (cathode) increases in mass due to the deposition of copper because of the reduction of copper ions. The blue colour of the solution diminishes due to the reduction of the blue copper ions. (b) (i) Oxygen concentration decreases (then increases in some areas) NB Cause: Effect (ii) Cause: Lot of oxygen at surface At the surface there is a lot of photosynthetic organisms producing oxygen and high partial pressure of oxygen above water results in oxygen dissolving. [1 mark] : As depth increases, less photosynthetic organisms but many organisms respiring and using oxygen causing it to decrease.[1 mark] (At great depths water from Arctic and Antarctic regions feeds in. This water is high on oxygen hence concentration increases slowly.) <u>Work of Early Scientists</u> <u>Galvani</u> Galvani discovered effect of static charge and electric currents on muscles and nerves in frogs. Frog muscles contracted when the spinal cord was connected to iron with a copper hook. Twitching occurred when the nerve was touched with different metals. Conclusion was the tissues contain an electric fluid called <i>“animal electricity”</i> as no external source of electricity was involved. Believed the brain created the fluid and nerves conducted the fluid to muscles which were stimulated by it. <u>Volta</u> Repeated and extended his experiments with frogs legs based on extending the idea of whether the electricity was in the muscle or arose from the metals connected by moist salty flesh of frogs leg. He tested different metals and the twitching got stronger as the similarity between the metals decreased. Described Galvani’s “animal electricity” as “metal electricity”. So concluded that “life, tissue or a vital force” not responsible but it was the metals. The	2 Marks - Greater than 1.02V with working 1 Mark - Correct answer with equations but no reference to greater than OR greater than 1.02V but no working OF incorrect answer but correct use of equations. 2 Marks - 2 valid observations with explanations. 1 Mark - 1 valid observation with explanation OR 2 valid observations with no explanation. 1 Mark – Identifies the trend correctly 2 Marks - Full working and correct answer 3-4 Marks Outlines the work of Galvani, Volta, Davy and faraday in increasing our understanding of electron transfer reactions AND provides features and characteristics of the impact of this work on society. 2 marks States the work of at least two of Galvani, Volta, Davy and faraday increasing our understanding of electron transfer reactions OR provides features and characteristics of the impact of this work on society. 1 mark States the work of at least one of Galvani, Volta, Davy and faraday increasing our understanding of electron transfer reactions
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salty environment from the frogs led connected the metals. Metal **contact theory** important lead into electrochemistry.

Davy

Developed the voltaic pile to create a bigger battery and electrolysed water. Electrolysed molten KOH and NaOH to produce K and Na. after realizing aqueous solutions only produced hydrogen and oxygen gas. Went on to isolate Sr, Ca, Mg, Ba Electric current through molten salts, resulting in **reduction reactions** to produce metals. Electrolyte used regulates the **substance formed** at each electrode. Compounds contained charges particles which **conduct electricity** by moving to the opposite charged electrode. An electric force holds compounds together

Faraday

Electrolytic reactions leading to the development of the Faraday's Laws of Electrolysis. Amount of an element produced (**quantitative analysis**) by electrolysis dependent on quantity of electricity passed through the circuit, the atomic mass, the element and the valency. Quantitative laws supported the concept at the time **that electricity was a particle (electron)**. Electrolyte theory that salts break up and the resulting charged particles move to the opposite electrodes. Terms such as anode, cathode, anion, cation, electrode and electrolysis which helped his descriptions of the experiments he carried out.

Impact on society

The above shows the work done by each scientist in assisting in an understanding of electron transfer reactions. With this knowledge came significant impacts on society such as:

- Portable forms of electricity enabling operation of devices such as torches.
- Isolation of metals through electrolysis eg copper to be used in electrical wiring
- Transport is electricity dependent, as is heating lighting, entertainment

(d)

(i) Set up test tubes according to the following table

Test tube #	Metal	Vol of Soln	Soln
1	0.5cm ² Mg	5mL	0.1 mol/L HCl
2	0.5cm ² Fe	5mL	0.1 mol/L HCl
3	0.5cm ² Mg	5mL	1.0 mol/L HCl
4	0.5cm ² Fe	5mL	1.0 mol/L HCl
5	0.5cm ² Mg	5mL	0.1 mol/L CH ₃ COOH
6	0.5cm ² Fe	5mL	0.1 mol/L CH ₃ COOH
7	0.5cm ² Mg	5mL	1.0 mol/L CH ₃ COOH
8	0.5cm ² Fe	5mL	1.0 mol/L CH ₃ COOH
9	0.5cm ² Mg	5mL	0.1 mol/L NaCl
10	0.5cm ² Fe	5mL	0.1 mol/L NaCl
11	0.5cm ² Mg		1.0 mol/L NaCl

2 marks

Sketches in general terms the relevant procedure identifying all conditions [1]
AND shows results [1]

12	0.5cm ² Fe		1.0 mol/L NaCl
13	0.5cm ² Mg		Distilled water
14	0.5cm ² Fe		Distilled water

Results

Test Tube	Results
1	4
2	3
3	5
4	4
5	3
6	2
7	4
8	3
9	2
10	2
11	3
12	3
13	2
14	2

Scale of 0-5 of corrosion, measured by recording appearance and size of metal after all metals being submersed for exactly 24 hours.

0 = no reaction

1 = very small amount corrosion

2 = small amount of corrosion

3 = moderate corrosion

4 = significant corrosion

5 = high corrosion

(ii)

Identify: Acidic solutions

Assess: Acidic solutions are corrosive to the skin.

Control: Use a dropper to carefully transfer the acid to the test tube to minimise splashing upon transfer. Wear gloves as a final protection measure.

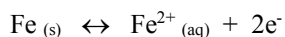
(iii)

In the experiment metal corrosion was significantly accelerated in the acidic environment as compared to the neutral environments.

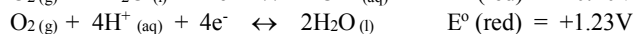
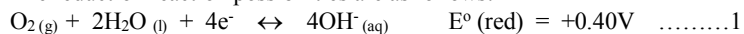
The higher the acid concentration, the greater the rate of corrosion the metal. The more reactive Mg corroded faster than the Fe.

Iron corrosion can be used to explain why this is the case.

Iron oxidises according to the following equation:



The reduction reaction possibilities are as follows:



.....2

The acidic environment (eg presence of SO_{2(g)} due to human activity) provides a more spontaneous reaction. Also, the H⁺ ions from reaction 2 react with the OH⁻ ions from reaction 1. This drives the first reaction to the right, therefore increasing the E^o of equation 1.

Full RA [1]

Relates results of experiment [1]

Provides correct equations [1]

Correct explanation [1]